Son 3 puntos total

Rotaciones =

$$P' = \underbrace{R(\theta_2)}_{M_2} \cdot \{\underbrace{R(\theta_1)}_{M_1} \cdot P\}$$ 0,2

Matriz elemental de Rotación

$$\begin{bmatrix} \cos\theta_1 & -\operatorname{sen}\theta_1 & 0 \\ \operatorname{sen}\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P' = \{\underbrace{R(\theta_2) \cdot R(\theta_1)}_{M}\} \cdot P$$

$$R(\theta_2) \cdot R(\theta_1) = R(\theta_2+\theta_1) \Leftrightarrow P' = R(\theta_2+\theta_1) \cdot P$$

tenemos esta igualdad como verdad!! M_1 y M_2 matrices de Rotación

$$\Rightarrow \overset{M_2}{\begin{bmatrix} \cos\theta_2 & -\operatorname{sen}\theta_2 & 0 \\ \operatorname{sen}\theta_2 & \cos\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}} \cdot \overset{M_1}{\begin{bmatrix} \cos\theta_1 & -\operatorname{sen}\theta_1 & 0 \\ \operatorname{sen}\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$$ 0,5

$$M \quad M_2 \cdot M_1 = R(\theta_2) \cdot R(\theta_1) = R(\theta_2+\theta_1)$$ 0,2

Calcular componente Q_{ij} de $R(\theta_2+\theta_1)$

0,1 $Q_{11} = \cos(\theta_2) \cdot \cos(\theta_1) - \operatorname{sen}(\theta_2) \cdot \operatorname{sen}(\theta_1) + 0 \cdot 0 = \cos(\theta_2) \cdot \cos(\theta_1) - \operatorname{sen}(\theta_2) \cdot \operatorname{sen}(\theta_1)$

0,1 $Q_{12} = -(\cos(\theta_2)\operatorname{sen}(\theta_1) + \operatorname{sen}(\theta_2) \cdot \cos(\theta_1)) + 0 \cdot 0 = -(\cos(\theta_2)\operatorname{sen}(\theta_1) + \operatorname{sen}(\theta_2) \cdot \cos(\theta_1))$

0,1 $Q_{13} = \cos\theta_2 \cdot 0 + -\operatorname{sen}\theta_2 \cdot 0 + 0 \cdot 1 = 0$

0,1 $Q_{21} = \operatorname{sen}\theta_2 \cdot \cos\theta_1 + \cos\theta_2 \cdot \operatorname{sen}\theta_1 + 0 \cdot 0 = \operatorname{sen}\theta_2 \cdot \cos\theta_1 + \cos\theta_2 \cdot \operatorname{sen}\theta_1$

0,1 $Q_{22} = \operatorname{sen}\theta_2 \cdot -\operatorname{sen}\theta_1 + \cos\theta_2 \cdot \cos\theta_1 + 0 \cdot 0 = -\operatorname{sen}\theta_2 \cdot \operatorname{sen}\theta_1 + \cos\theta_2 \cdot \cos\theta_1$

0,1 $Q_{23} = \operatorname{sen}\theta_2 \cdot 0 + \cos\theta_2 \cdot 0 + 0 \cdot 1 = 0$

0,1 $Q_{31} = 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 = 0$

0,1 $Q_{32} = 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 = 0$

0,1 $Q_{33} = 0 \cdot 0 + 0 \cdot 0 + 1 \cdot 1 = 1$

Matriz M resultante 1

$$\begin{bmatrix} Q_{11} & Q_{12} & Q_{13} \\ Q_{21} & Q_{22} & Q_{23} \\ Q_{31} & Q_{32} & Q_{33} \end{bmatrix} = \begin{bmatrix} \cos(\theta_2) \cdot \cos(\theta_1) - \operatorname{sen}(\theta_2) \cdot \operatorname{sen}(\theta_1) & -(\cos(\theta_2)\operatorname{sen}(\theta_1) + \operatorname{sen}(\theta_2) \cdot \cos(\theta_1)) & 0 \\ \operatorname{sen}(\theta_2) \cdot \cos(\theta_1) + \cos(\theta_2) \cdot \operatorname{sen}(\theta_1) & -\operatorname{sen}(\theta_2)\operatorname{sen}(\theta_1) + \cos(\theta_2) \cos(\theta_1) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$P^T = [2.4, 5, 1]$ encontrar P' Aplicando M. Con los angulos $\theta_1 = 30°$ y $\theta_2 = -45°$

$$\begin{bmatrix} \cos(\theta_2) \cdot \cos(\theta_1) - \operatorname{sen}(\theta_2) \cdot \operatorname{sen}(\theta_1) & -(\cos(\theta_2)\operatorname{sen}(\theta_1) + \operatorname{sen}(\theta_2) \cdot \cos(\theta_1)) & 0 \\ \operatorname{sen}(\theta_2) \cdot \cos(\theta_1) + \cos(\theta_2) \cdot \operatorname{sen}(\theta_1) & -\operatorname{sen}(\theta_2)\operatorname{sen}(\theta_1) + \cos(\theta_2)\cos(\theta_1) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2,4 \\ 5 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} (\cos(-45) \cdot \cos(30) - \operatorname{sen}(-45) \cdot \operatorname{sen}(30)) \cdot 2,4 - (\cos(-45)\operatorname{sen}(30) + \operatorname{sen}(-45) \cdot \cos(30)) \cdot 5 + 0 \cdot 1 \\ \operatorname{sen}(-45)\cos(30) + \cos(-45)\operatorname{sen}(30) \cdot 2,4 + (-\operatorname{sen}(-45)\operatorname{sen}(30) + \cos(-45)\cos(30)) \cdot 5 + 0 \cdot 1 \\ 0 \cdot 2,4 + 0 \cdot 5 + 1 \cdot 1 \end{bmatrix} =$$

$$\begin{bmatrix} (0,525321 \cdot 0,154251 - (-0,80903 \cdot -0,98803)) 2,4 \quad -0,525321 \cdot -0,988031 + (-0,850903 \cdot 0,154251)) 5 + 0 \\ (0,80903 \cdot 0,154251 + 0,525321 \cdot -0,98803) \cdot 2,4 + (-0,80903 \cdot -0,988031 + (0,525321 \cdot 0,154251)) 5 + 0 \\ 0 + 0 + 1 \end{bmatrix}$$

$$\begin{bmatrix} -1,72395 - 1,9389 + 0 \\ -0,946169 + 4,40189 + 0 \\ 0 + 0 + 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -3,69285 \\ 3,45572 \\ 1 \end{bmatrix} = P'$$

0,2